Workshop: Explainable AI (XAI)

Mastering Explainability

2026.04.30

Seung Hyun Han
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- Preliminary Knowledge
  - Understanding Machine Learning
  - Black-Box Model
  - Visualization vs XAI
- Decision Tree
  - Visualizing Decision Tree
  - Feature Importance
  - Partial Dependence Plot
  - Xgboost
- Surrogate Analysis
  - LIME
  - SHAP

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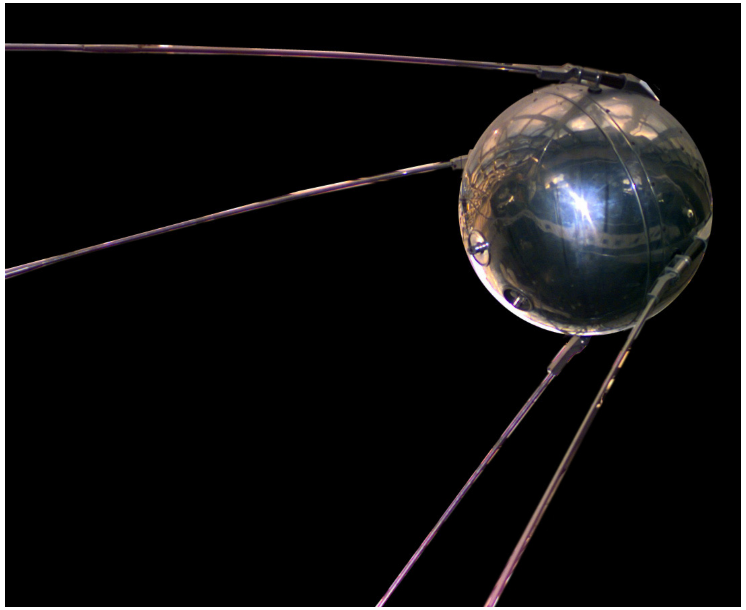
Introduction

- DARPA (Defence Advanced Research Projects Agency)
  - Sputnik Crisis (Спутниковый кризис) 1957.10.4



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  - Dwight David Eisenhower → James Rhyne Killian (former MIT president)
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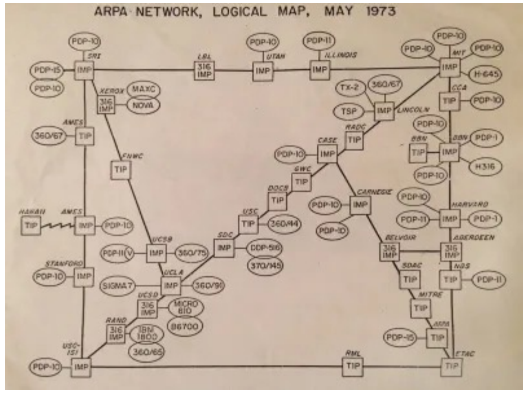

  - ARPnet(1969) www based internet (1990.12.20)

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- 2016.8.10
  - DARPA-BAA-16-53

Broad Agency Announcement

Explainable Artificial Intelligence (XAI)

DARPA-BAA-16-53

August 10, 2016

Google DeepMind Challenge Match

Google DeepMind Challenge Match

Google DeepMind Challenge Match

period	2016 March 9th - March 15th
host	Google
progress pla ce	Four Seasons Hotel Seoul (Republic of Korea , Seoul Metropolitan City Jongno- gu)
Participant	AlphaGo vs Lee Sedol 9

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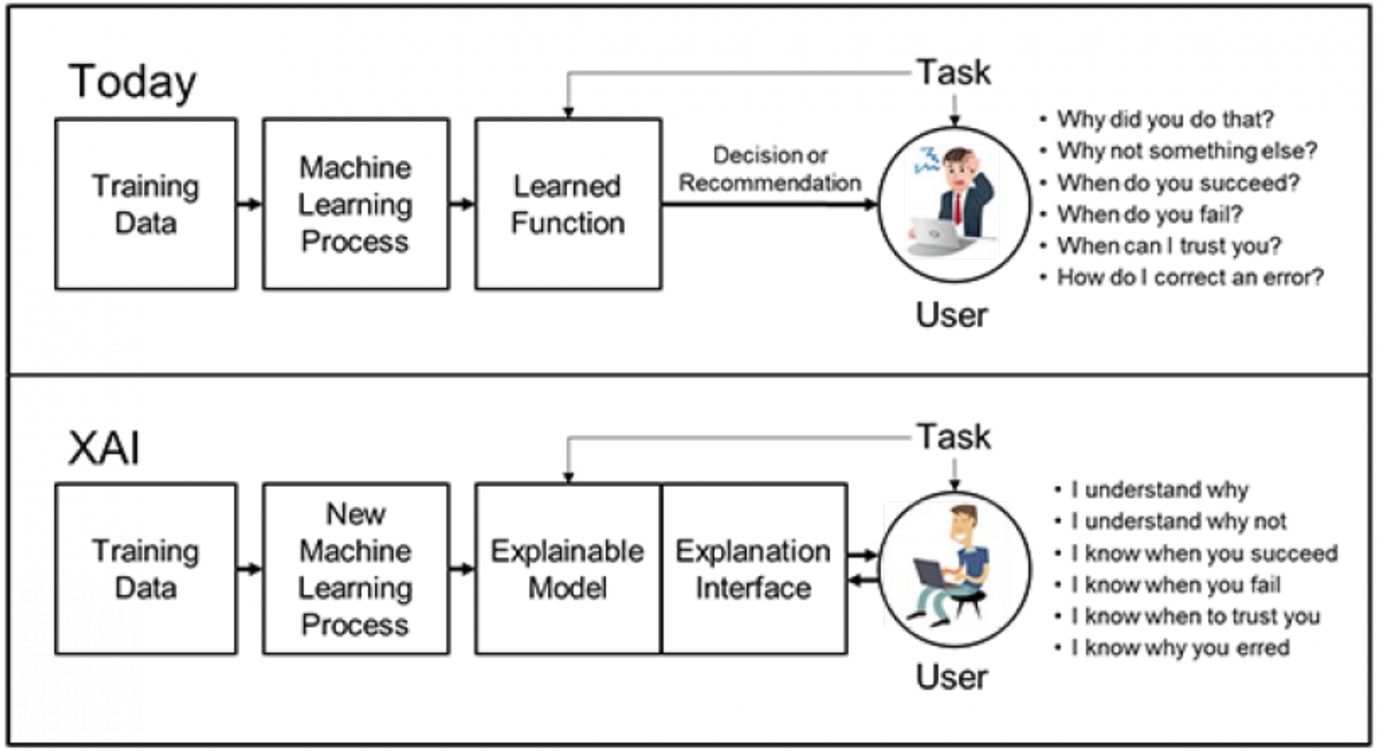

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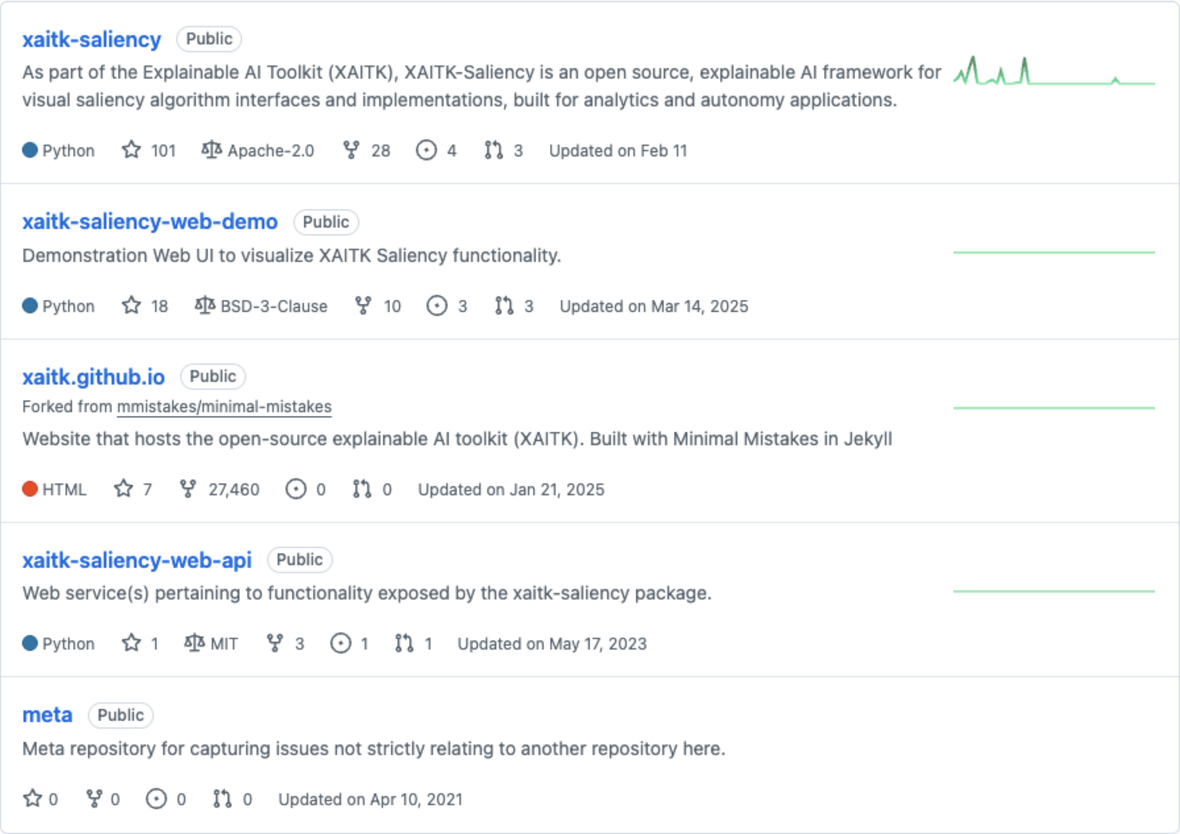
  - Outcome : https://xaitk.org/



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  - Shortliffe, Edward H., and Bruce G. Buchanan. "A model of inexact reasoning in medicine." *Mathematical biosciences* 23.3-4 (1975): 351-379.

ABSTRACT

Medical science often suffers from having so few data and so much imperfect knowledge that a rigorous probabilistic analysis, the ideal standard by which to judge the rationality of a physician's decision, is seldom possible. Physicians nevertheless seem to have developed an ill-defined mechanism for reaching decisions despite a lack of formal knowledge regarding the interrelationships of all the variables that they are considering. This report proposes a quantification scheme which attempts to model the inexact reasoning processes of medical experts. The numerical conventions provide what is essentially an approximation to conditional probability, but offer advantages over Bayesian analysis when they are utilized in a rule-based computer diagnostic system. One such system, a clinical consultation program named MYCIN, is described in the context of the proposed model of inexact reasoning.

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  - Van Lent, Michael, William Fisher, and Michael Mancuso. "An explainable artificial intelligence system for small-unit tactical behavior." *Proceedings of the national conference on artificial intelligence*. Menlo Park, CA; Cambridge, MA; London; AAAI Press; MIT Press; 1999, 2004.

Abstract

As the artificial intelligence (AI) systems in military simulations and computer games become more complex, their actions become increasingly difficult for users to understand. Expert systems for medical diagnosis have addressed this challenge though the addition of explanation generation systems that explain a system's internal processes. This paper describes the AI architecture and associated explanation capability used by Full Spectrum Command, a training system developed for the U.S. Army by commercial game developers and academic researchers.

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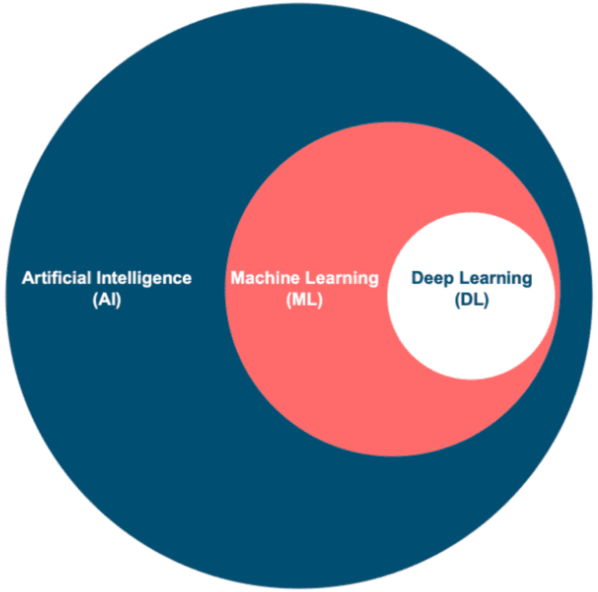
Preliminary Knowledge : Re-visiting Machine Learning

- ML, DL and AI
  - AI: Capability of computer systems to perform tasks that typically require human intelligence.
  - ML: Algorithms learn patterns from data to make predictions or decisions without being explicitly programmed for each task.
  - DL: A multilayered neural networks to automatically learn hierarchical representations from large amounts of data.



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- ML vs DL
  - Data management, or so called, **Feature management**
  - Feature?

	Pclass	Age	Survived	Sex_male	Embarked_Q	Embarked_S	family
0	3	22	0	1	0	1	1
1	1	38	1	0	0	0	1
2	3	26	1	0	0	1	0
3	1	35	1	0	0	1	1
4	3	35	0	1	0	1	0

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    - A measurable non-target attribute(property)
  - Domain Knowledge

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  - Domain Knowledge
    - the specialized expertise and understanding of the specific field or subject

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0							
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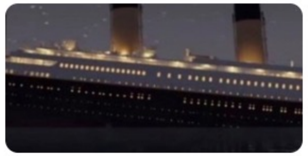
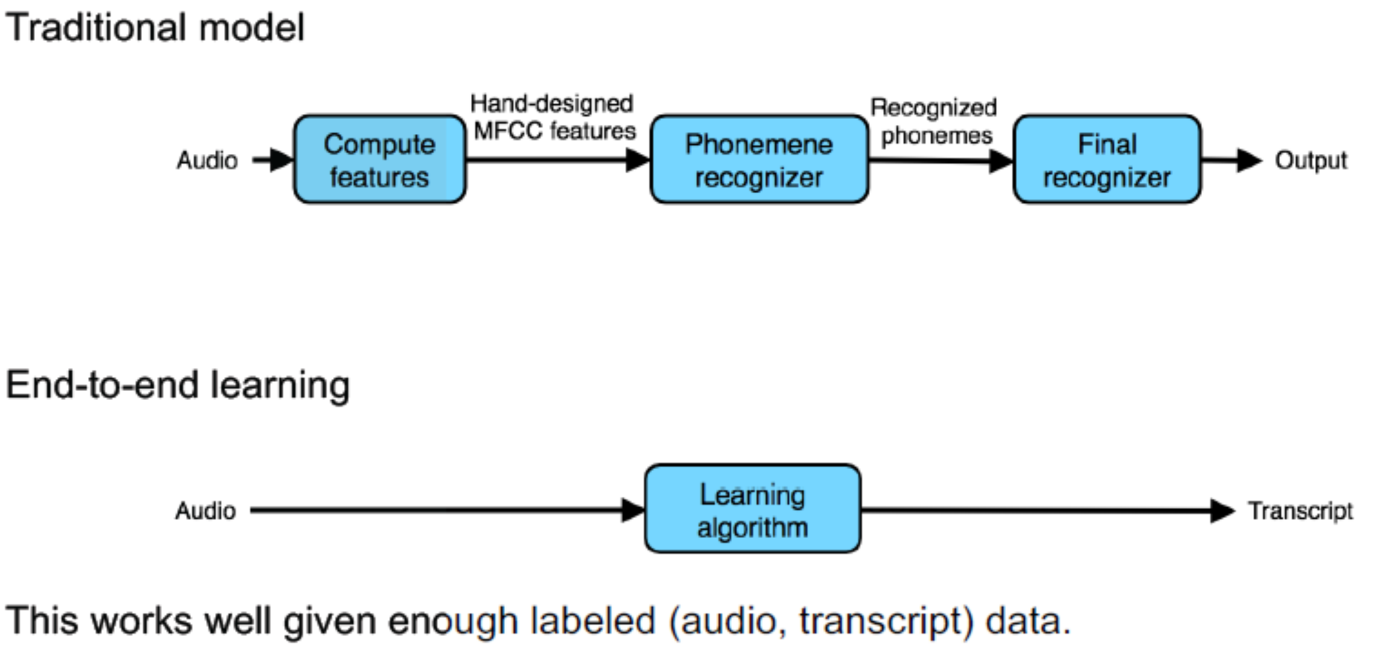
Titanic - Machine Learning from Disaster

Start here! Predict survival on the Titanic and get familiar with ML basics



Preliminary Knowledge : Re-visiting Machine Learning

- What is the Model?
  - ML/DL model is a mathematical or computational program that has learned **patterns from data** so it can make **predictions or decisions on new, unseen inputs**.
- DL with End-to-end?



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  - In social science, an "interpretability" is…
    - A process whereby a theory is transformed to a level that is understandable to its domain

Preliminary Knowledge : Re-visiting Machine Learning

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- DL with End-to-end?
- Black-box model?
  - In social science, an "interpretability" is…
    - A process whereby a theory is transformed to a level that is understandable to its domain
- Do you remember the previous lecture?
  - 3 major categories in contemporary research in AI?
    - **Domain adaptation**
    - Model lightening
    - **XAI**

Black Box Model

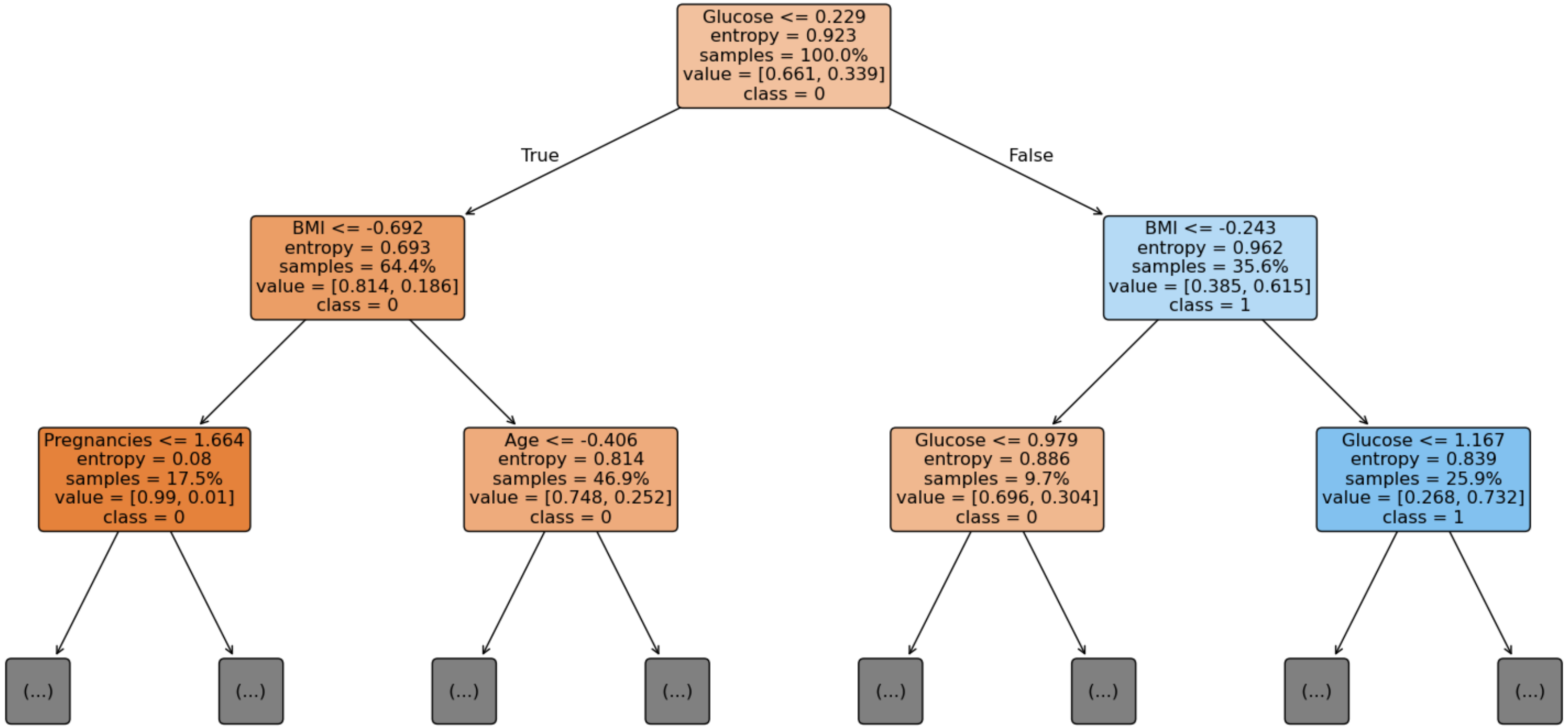
- Case study: diabetes issue in Pima Indians
  - Doctors collected native Pimas blood information

	id	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	576.000000	576.000000	576.000000	576.000000	576.000000	576.000000	576.000000	576.000000	576.000000	576.000000
mean	385.189236	3.866319	120.170139	68.421875	21.067708	82.133681	31.843403	0.479325	33.354167	0.338542
std	221.420334	3.388140	32.014163	20.080641	15.933359	115.444040	7.798820	0.343584	11.972995	0.473625
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	192.750000	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.244000	24.000000	0.000000
50%	385.500000	3.000000	116.000000	72.000000	23.000000	45.000000	32.050000	0.369000	29.000000	0.000000
75%	575.250000	6.000000	140.000000	80.000000	33.000000	130.000000	36.600000	0.637750	41.000000	1.000000
max	766.000000	17.000000	199.000000	114.000000	99.000000	846.000000	59.400000	2.420000	81.000000	1.000000

  - Draw a Tree

Black Box Model

- Case study: diabetes issue in Pima Indians
  - Doctors collected native Pimas blood information
  - Draw a Tree



Black Box Model

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  - Draw a Tree

First record from test dataset:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
0	-0.846722	2.276904	-0.918192	-0.318333	2.539073	-0.762752	0.511747	-0.781951

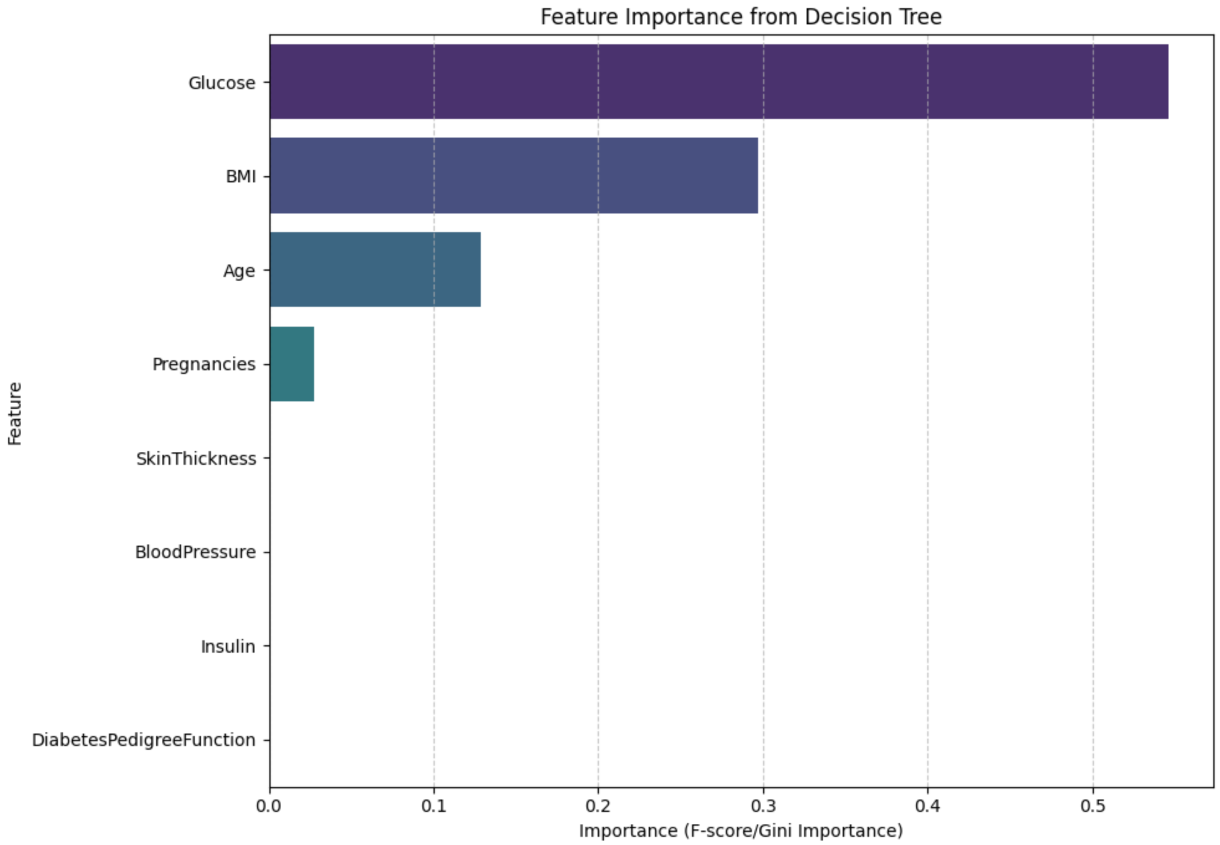
First record from original (unscaled) test dataset:

	id	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
0	258	1	193	50	16	375	25.9	0.655	24

Leaf node class counts (Class 0: 0.3889, Class 1: 0.6111)
Probability of Diabetes for this record: 0.6111

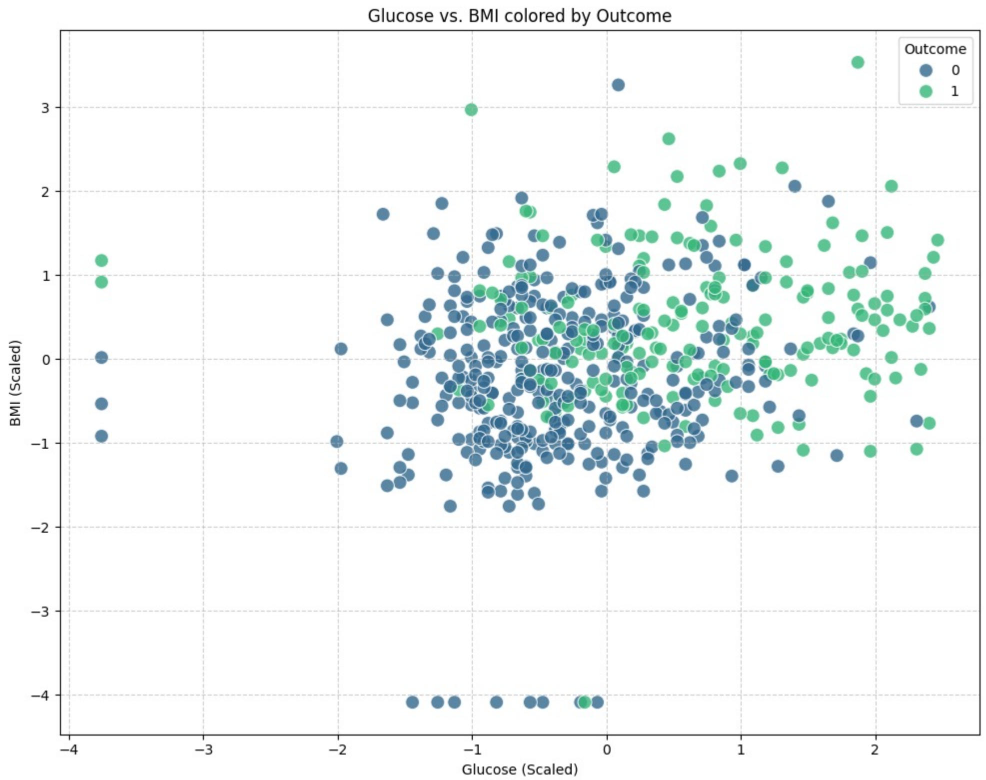
Black Box Model

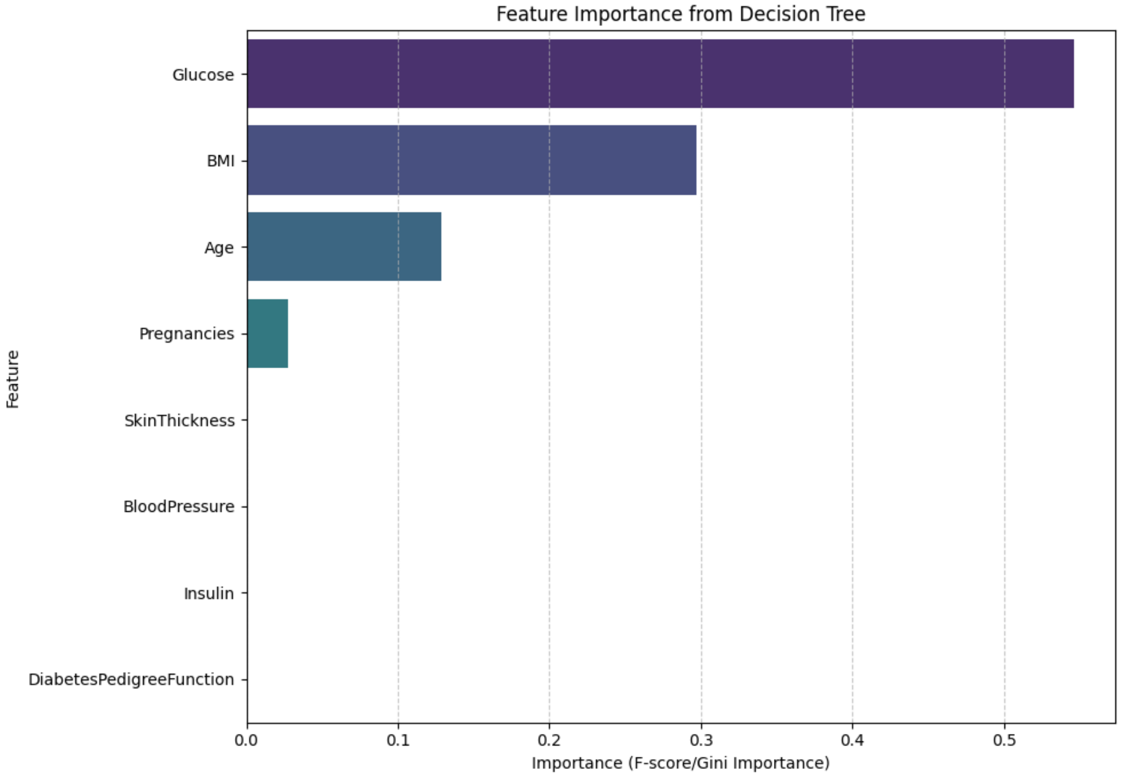
- Case study: diabetes issue in Pima Indians
  - Doctors collected native Pimas blood information
  - Draw a Tree
  - Draw a Feature Importance diagram



Visualization vs XAI

- Now it comes to have a question like…
  - Isn’t it a just visualization for certain models?
    - No, It looks like, but as long as the “interpretability” defines a ‘feature’s’ impact, it’s different from data visualization.
  - Back to Pima cases
    - Visualization & Feature Importance graph





Visualization vs XAI

- Now it comes to have a question like…
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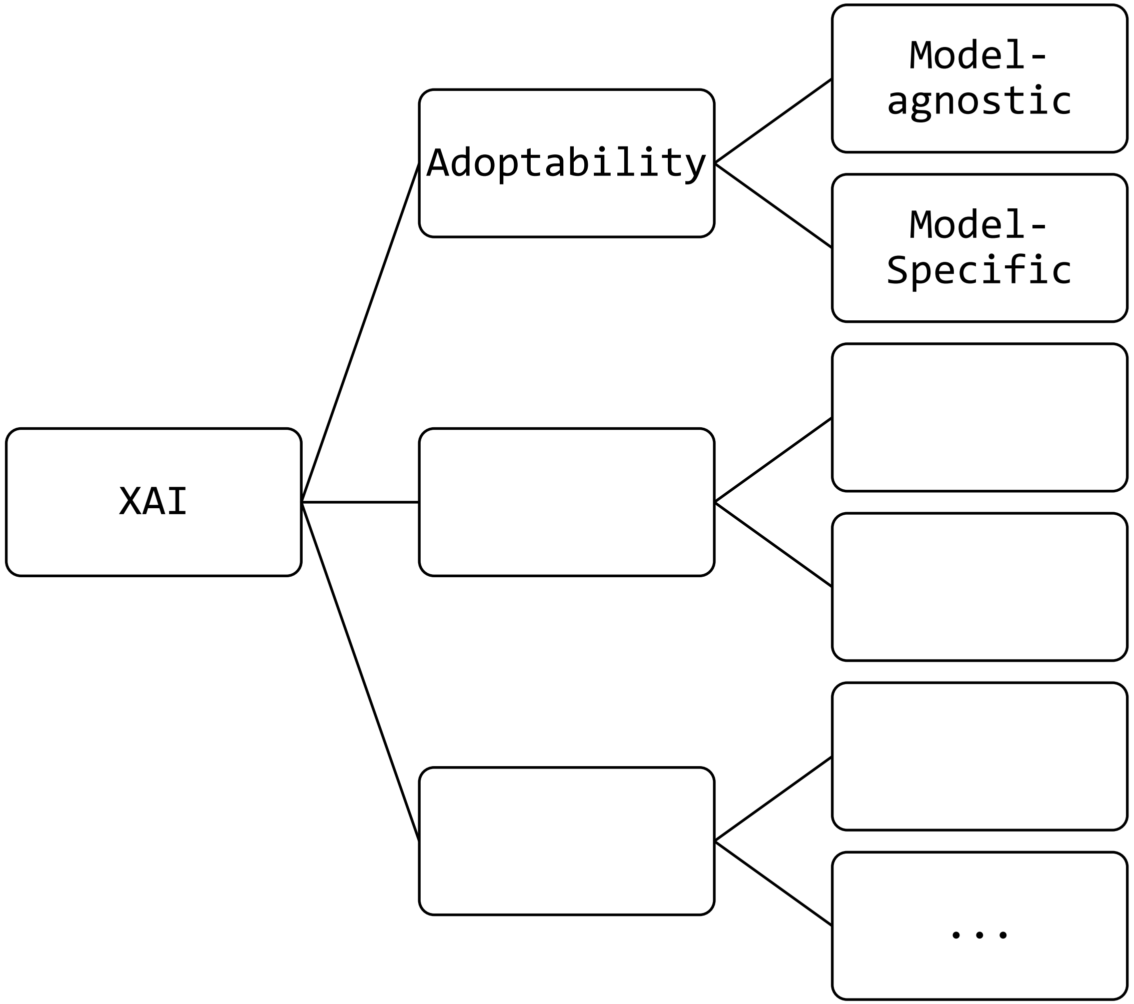
  - Back to Pima cases
    - Visualization & Feature Importance graph
    - Visualization
      - Tools such as histograms, scatter plots, and line graphs that show distributions, trends, and correlations in the data.
      - **Understand the data**
    - XAI
      - explain how each input factor contributed to a specific prediction and in which direction
      - **Understand the model's decision-making logic**

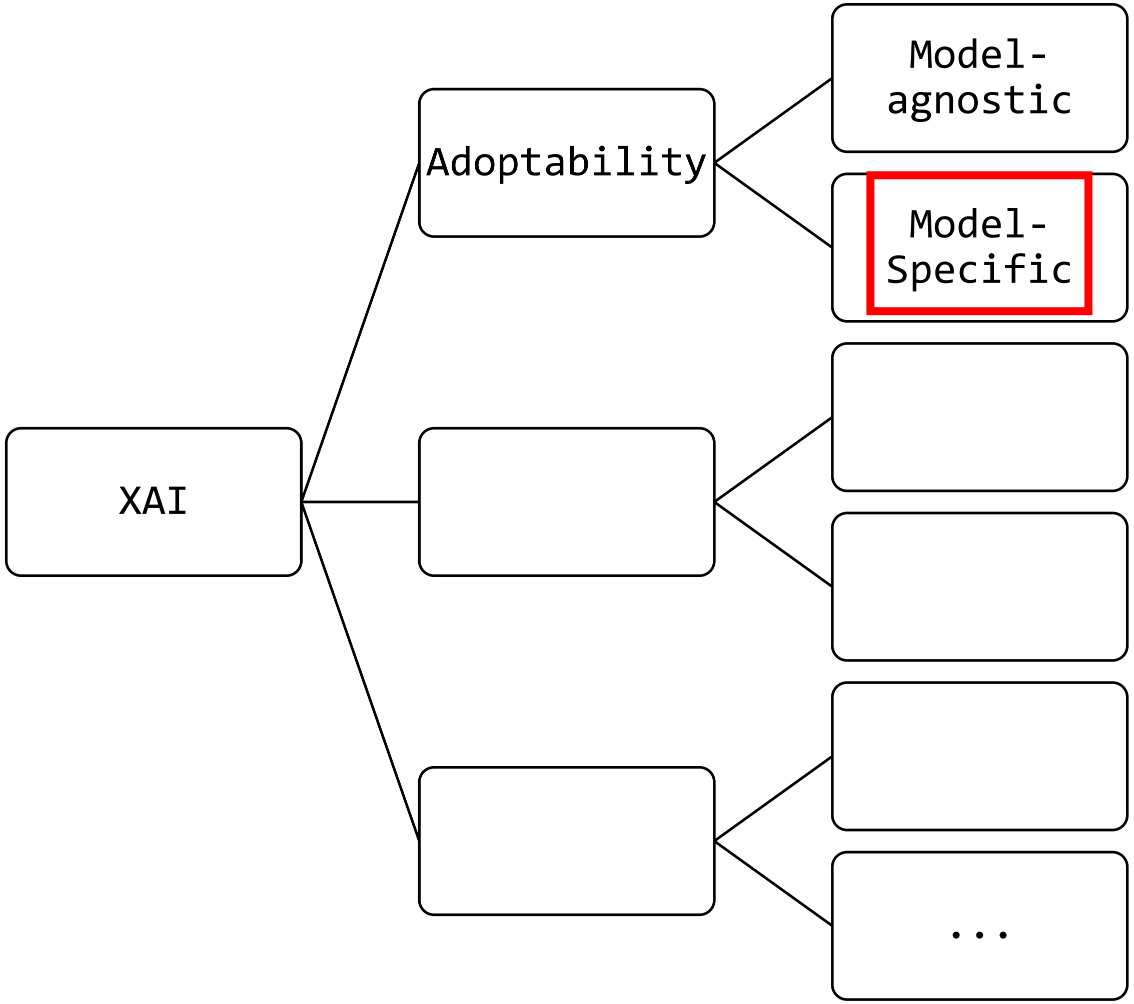
XAI Categorization



Does not use model's characteristic

Specialized on specific models only

XAI Categorization



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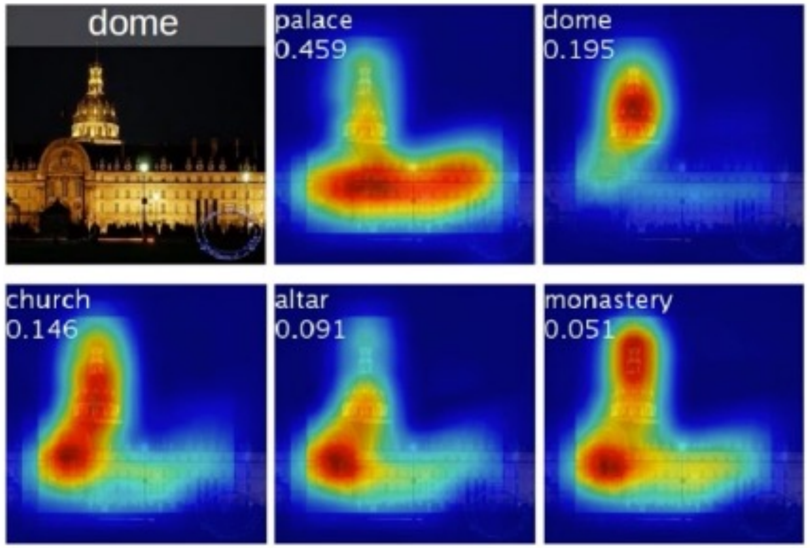
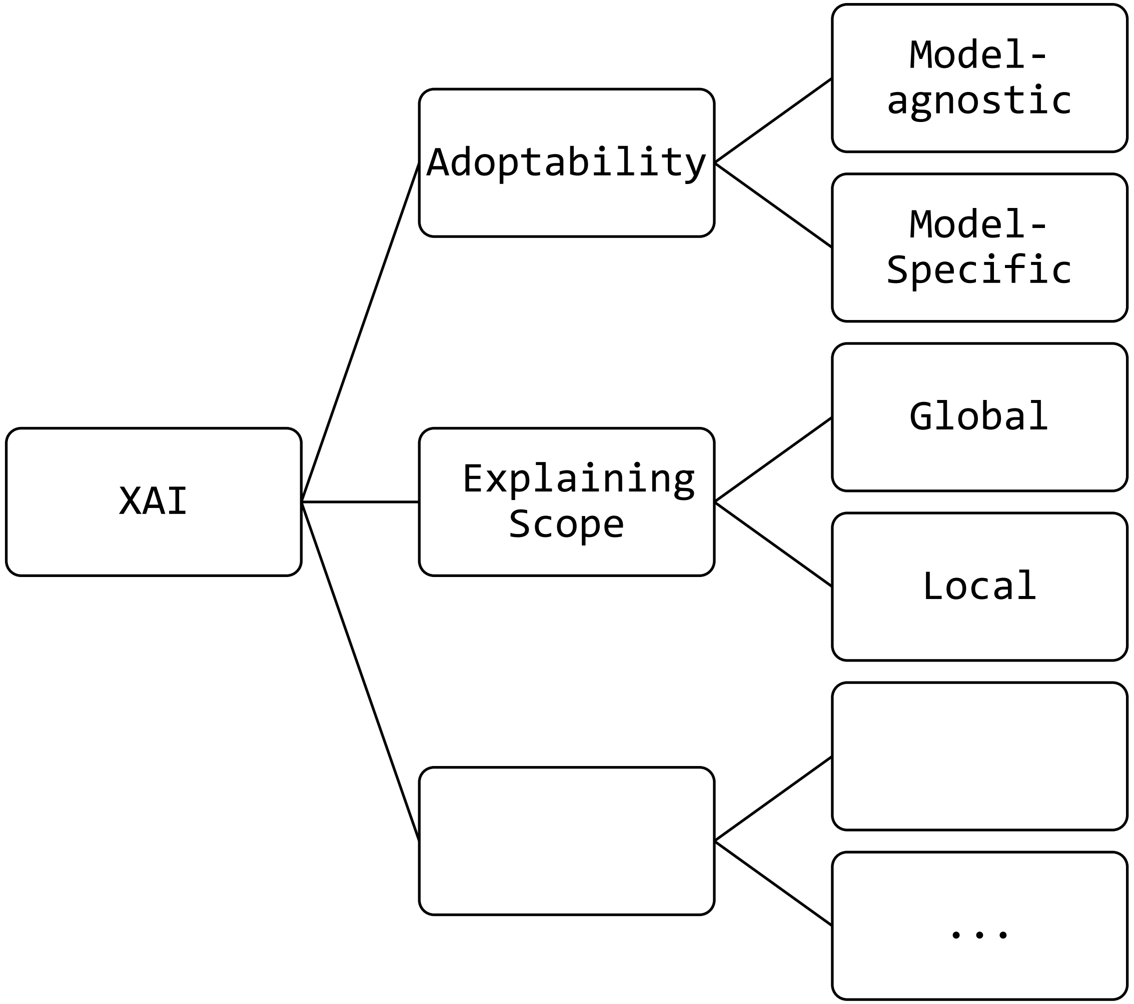


Class Activation Map(CAM)

XAI Categorization

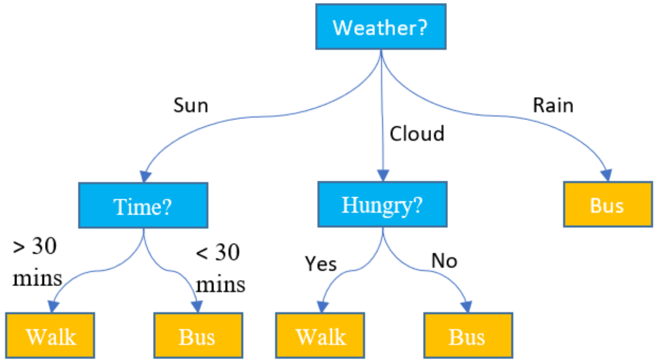


Does not use model’s characteristic

Specialized on specific models only

Explain all outcome of the model’s prediction
(e.g. Decision Tree)

Only Explains single outcome
(e.g. LIME)



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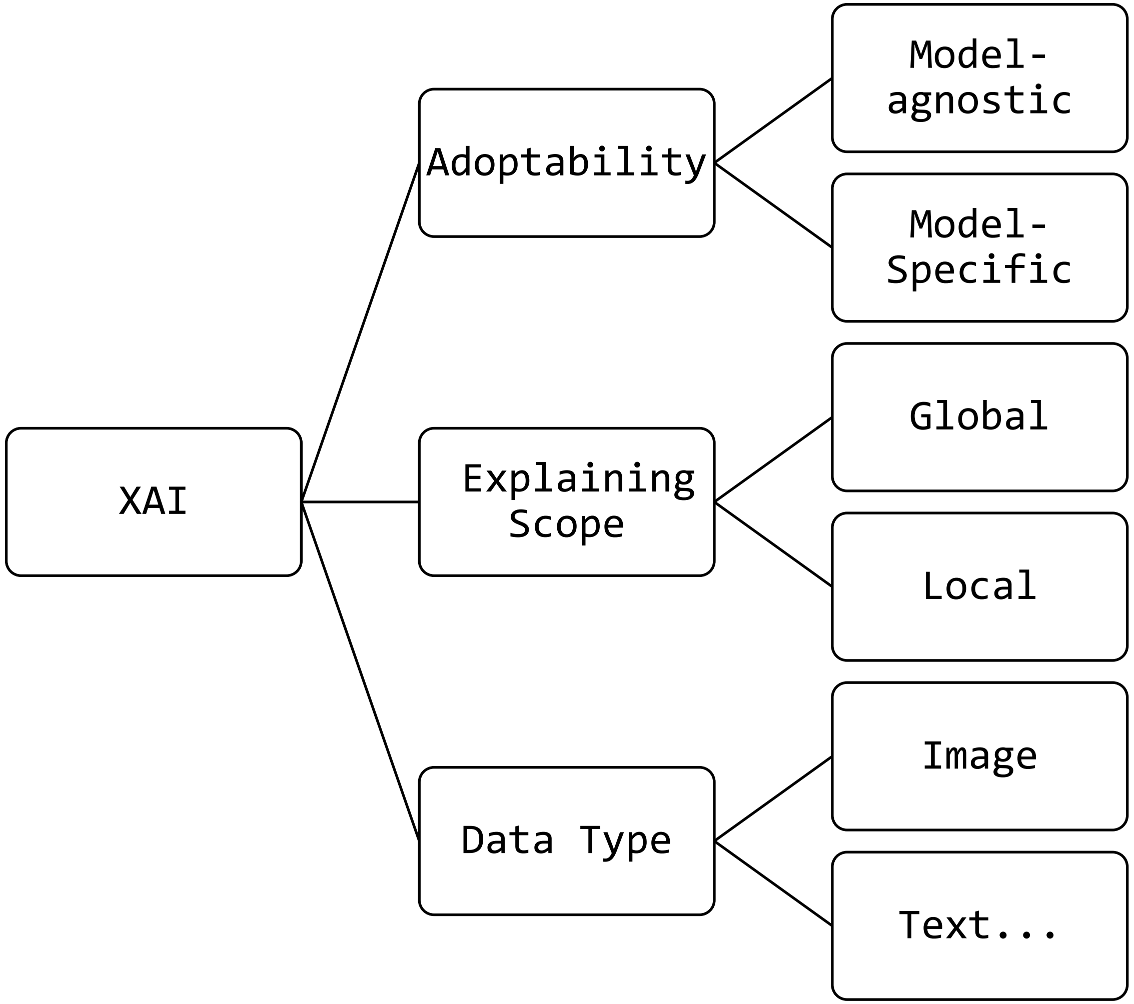

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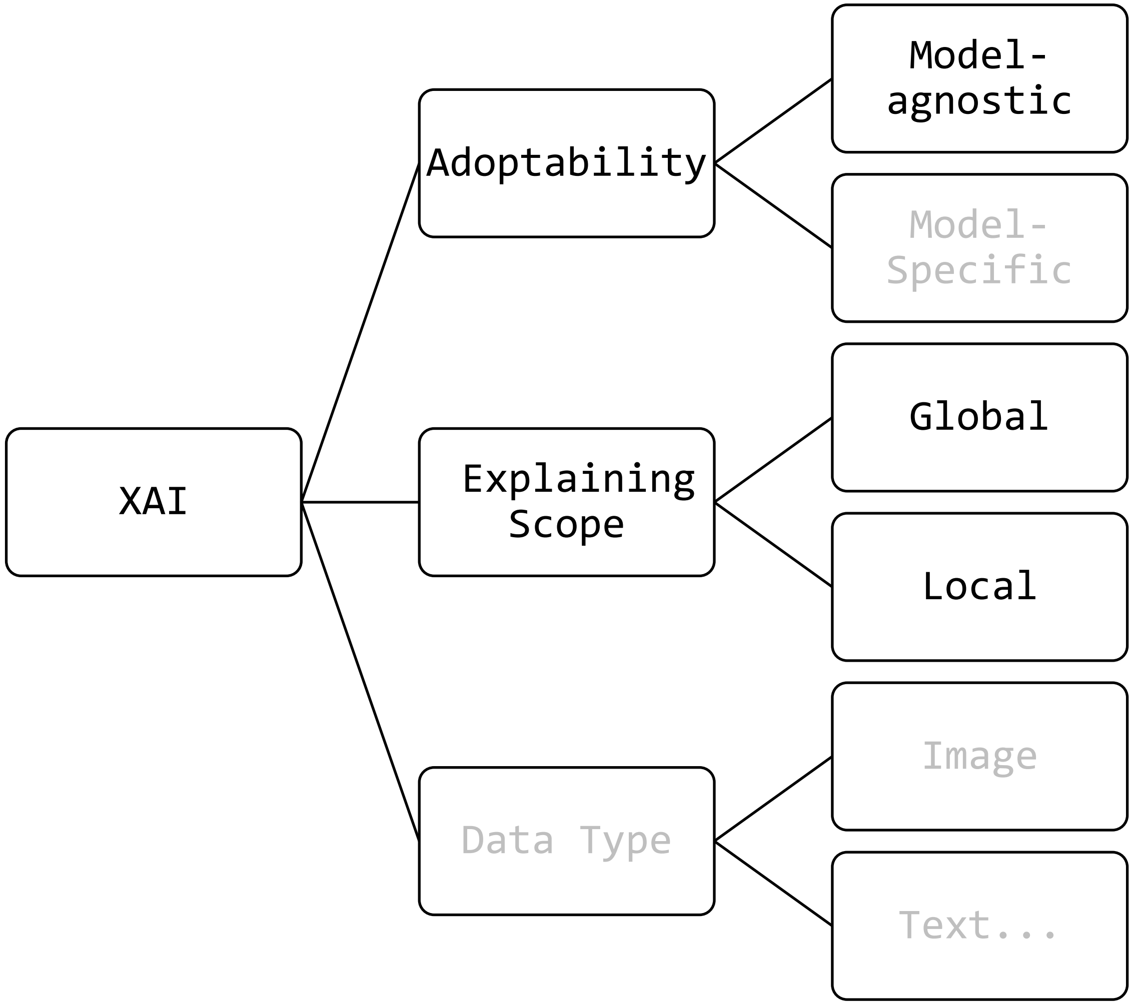

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LIME

- Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "" Why should i trust you?" Explaining the predictions of any classifier." *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. 2016.
- 33171 cited
- Core contribution: Insighted the Black-box modeling via Local Approximation

"Why Should I Trust You?" Explaining the Predictions of Any Classifier

Marco Tulio Ribeiro
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Seattle, WA 98105, USA
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Sameer Singh
University of Washington
Seattle, WA 98105, USA
sameer@cs.uw.edu

Carlos Guestrin
University of Washington
Seattle, WA 98105, USA
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ABSTRACT

Despite widespread adoption, machine learning models remain mostly black boxes. Understanding the reasons behind predictions is, however, quite important in assessing *trust*, which is fundamental if one plans to take action based on a prediction, or when choosing whether to deploy a new model. Such understanding also provides insights into the model, which can be used to transform an untrustworthy model or prediction into a trustworthy one.

In this work, we propose LIME, a novel explanation tech-

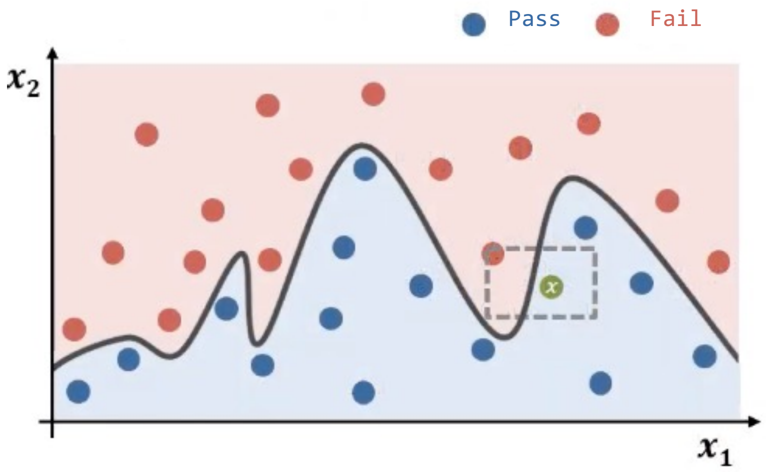
how much the human understands a model's behaviour, as opposed to seeing it as a black box.

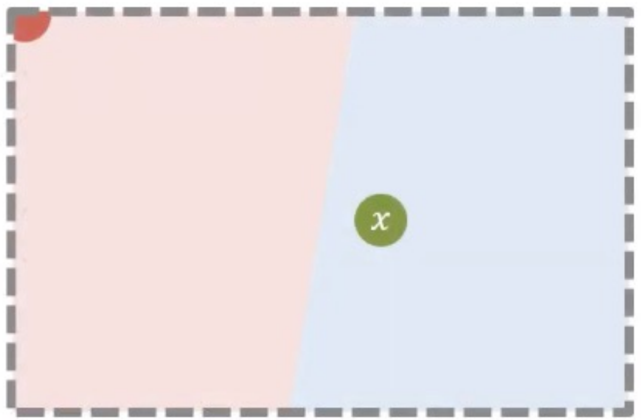
Determining trust in individual predictions is an important problem when the model is used for decision making. When using machine learning for medical diagnosis [6] or terrorism detection, for example, predictions cannot be acted upon on blind faith, as the consequences may be catastrophic.

Apart from trusting individual predictions, there is also a need to evaluate the model as a whole before deploying it "in the wild". To make this decision, users need to be confident

LIME (Local Interpretable Model-agnostic Explanations)

- Gives insights through *single observation* through Local Surrogate Model
  - Local Surrogate Model: Not entire dataset, but a single observation





x의 local area

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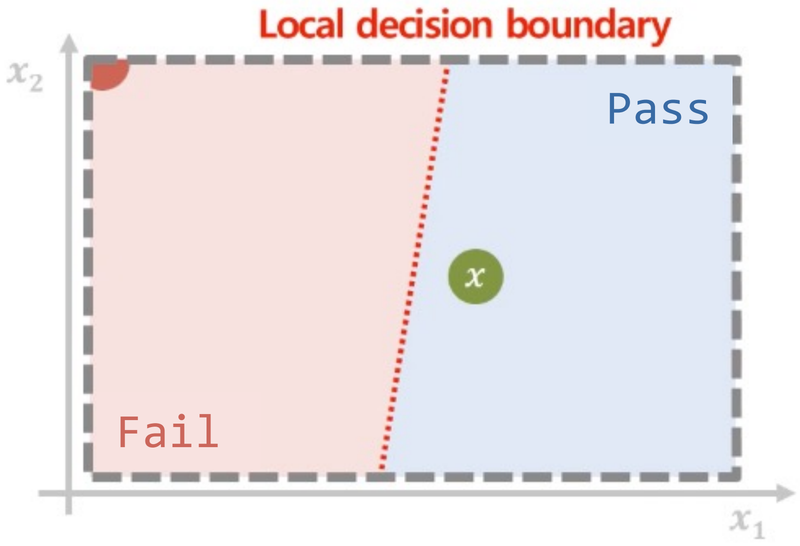

A solid explain model g against observation x

- A simple, interpretable model
- Tend toward target model

$$\xi(x) = \underset{g \in G}{argmin}\, \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

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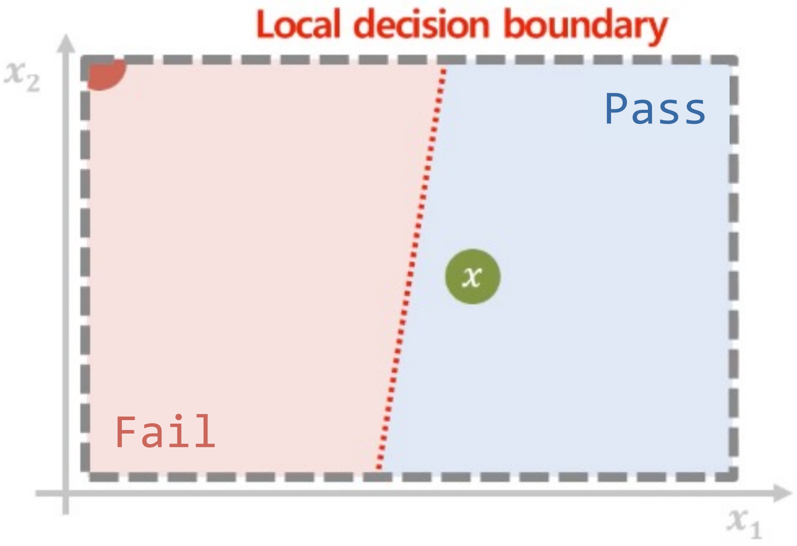
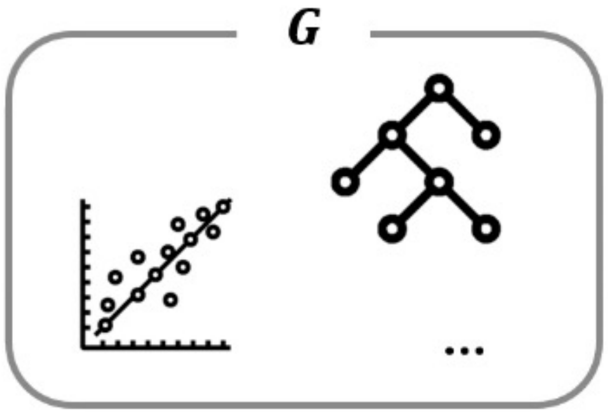


$$\xi(x) = \underset{\boxed{g \in G}}{argmin}\, \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

A set of good interpretability



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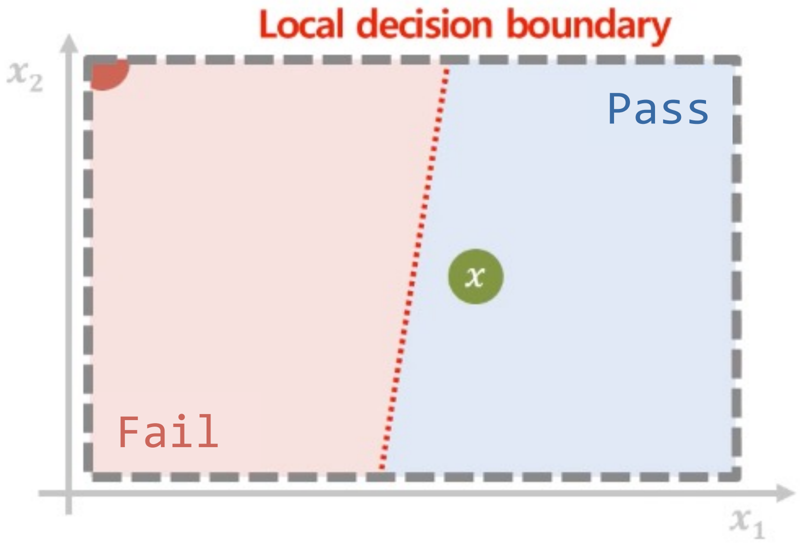


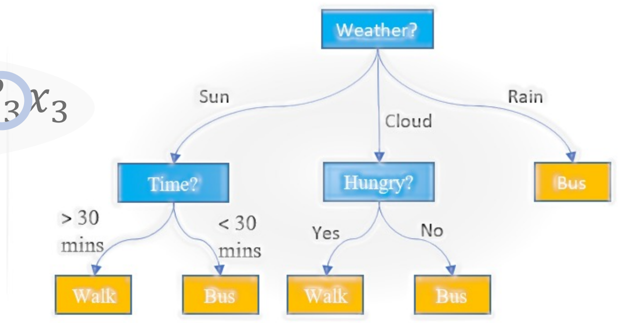
$$\xi(x) = \underset{g \in G}{argmin}\, \mathcal{L}(f, g, \pi_x) + \boxed{\Omega(g)}$$

Complexity of model g

Logistic Regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

Decision Tree



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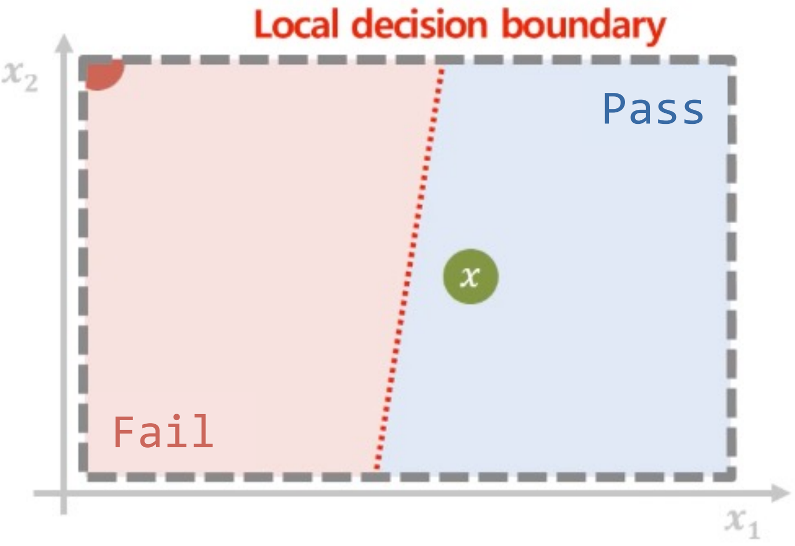
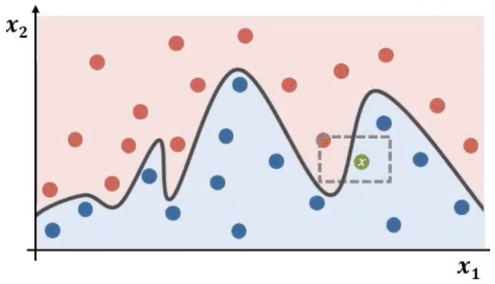


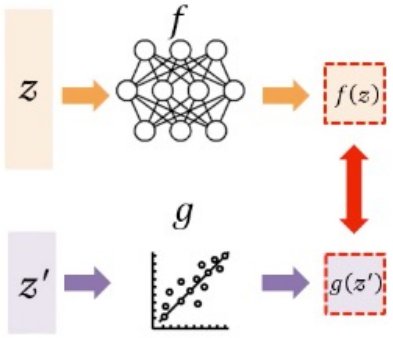


Complexed model

$$\xi(x) = \underset{g \in G}{argmin}\ \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

$$\mathcal{L}(f, g, \pi_x) = \sum_{z, z' \in Z} \pi_x(z)\left(f(z) - g(z')\right)^2$$

Dimensionality reduced z



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LIME (Local Interpretable Model-agnostic Explanations)

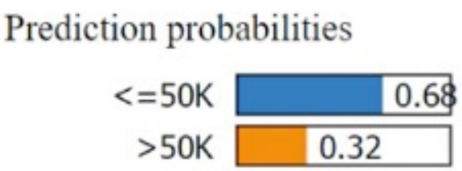
Minimize target model and explanation model

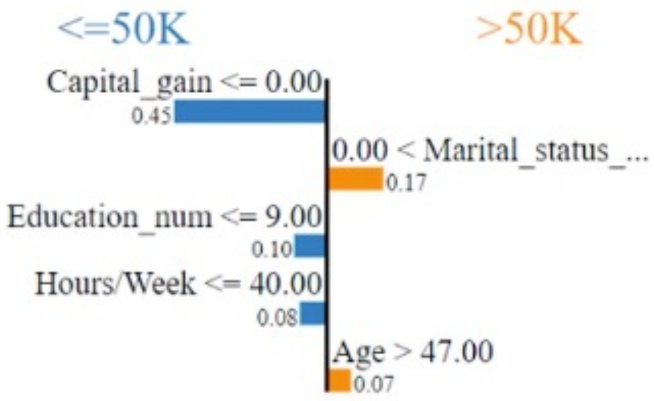
$$\xi(x) = \underset{g \in G}{argmin}\ \boxed{\mathcal{L}(f, g, \pi_x)} + \boxed{\Omega(g)}$$

Model complexity

LIME (Local Interpretable Model-agnostic Explanations)

- Binary Classification problem for Modern American income (50K $ up / down)
- LightGBM(Light Gradient Boosting Machine) Interpretability





LIME (Local Interpretable Model-agnostic Explanations)

- Contribution: an insights to the public with “white model”
- Widely used: both Neural Network & Random Forest
- Not using net-dataset decision boundary / but single observation decision boundary

Index

SHAP

- Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." *Advances in neural information processing systems* 30 (2017).
- 56619 cited
- Core contribution: Feature measurements from *Shapley values*

A Unified Approach to Interpreting Model Predictions

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Abstract

Understanding why a model makes a certain prediction can be as crucial as the prediction's accuracy in many applications. However, the highest accuracy for large modern datasets is often achieved by complex models that even experts struggle to interpret, such as ensemble or deep learning models, creating a tension between

SHAP

- Shapley Values
  - One of the well-known distributing strategy from Lloyd Shapley (1951)
  - A method based on game theory to fairly allocate rewards according to each player's contribution

Condition 1: The total allocated rewards = the total prize of the game ($\sum_i x_i = V$)

Condition 2: Players with higher contributions receive higher rewards

Two conditions for fair reward allocation

SHAP

- Shapley Values
  - A weighted average of each player's **Marginal Contributions**
  - Marginal contribution of a player:
    "For each possible team(subsets), how much the total payoff increases when **this player** joins that team."

SHAP

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  - A weighted average of each player's **Marginal Contributions**
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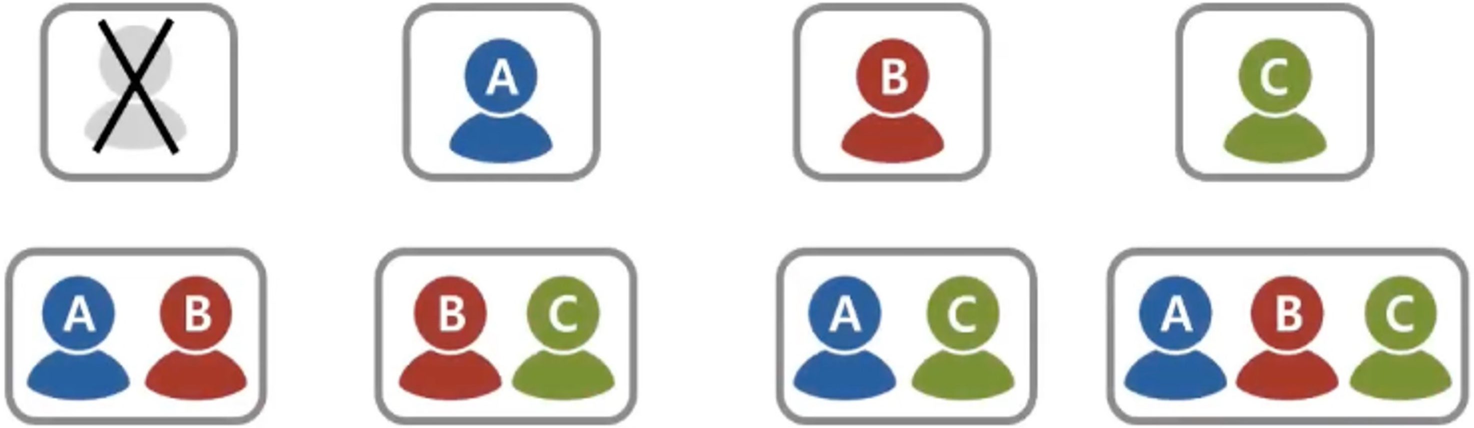
A's Shapley Value



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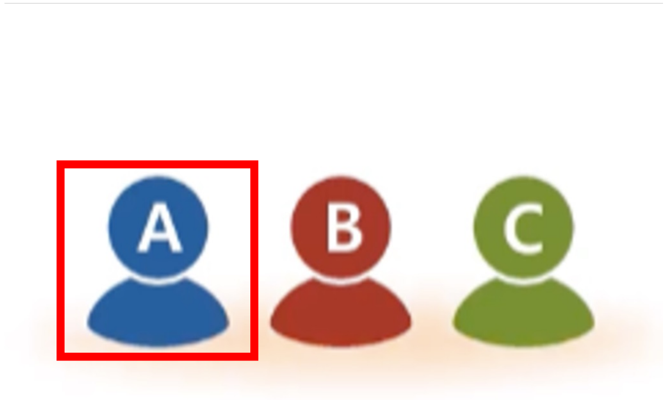

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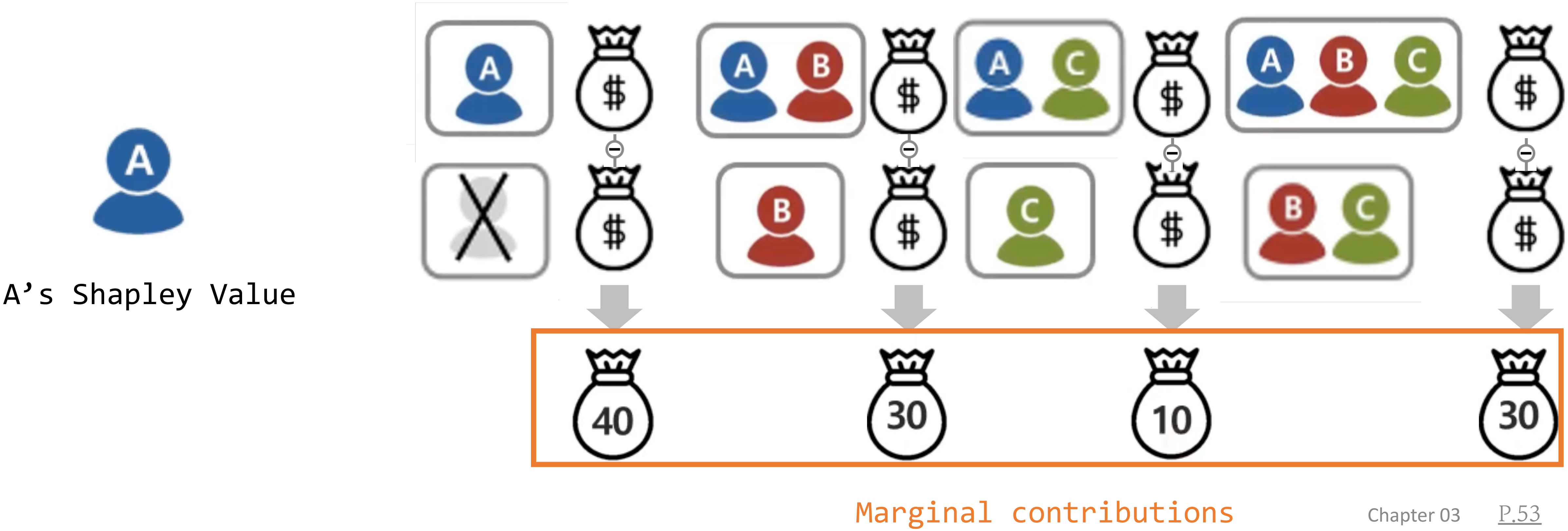
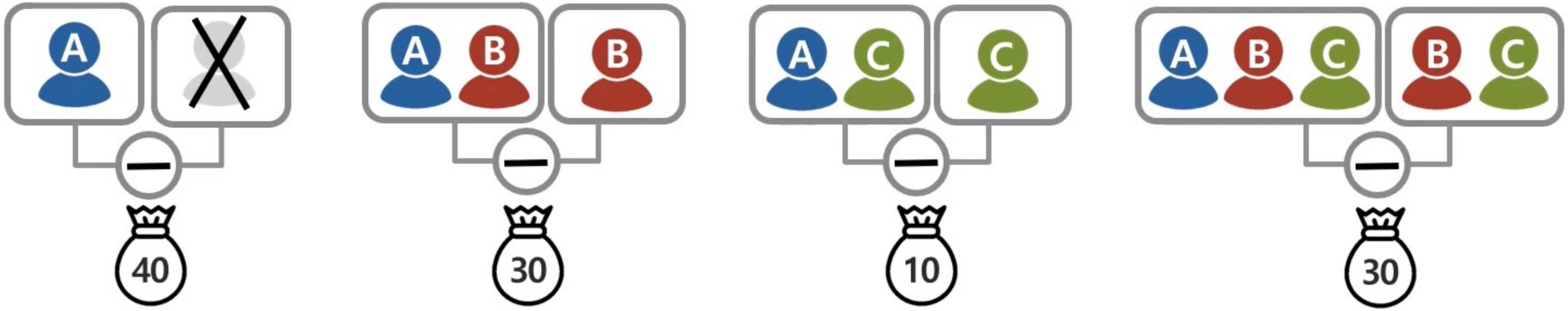


SHAP

- Shapley Values
  - A weighted average of each pla ... ations
  - Marginal contribution of a play
    “For each possible team(subset ... payoff increases when **this player** joins that team.”

$F = \{$  $\}$



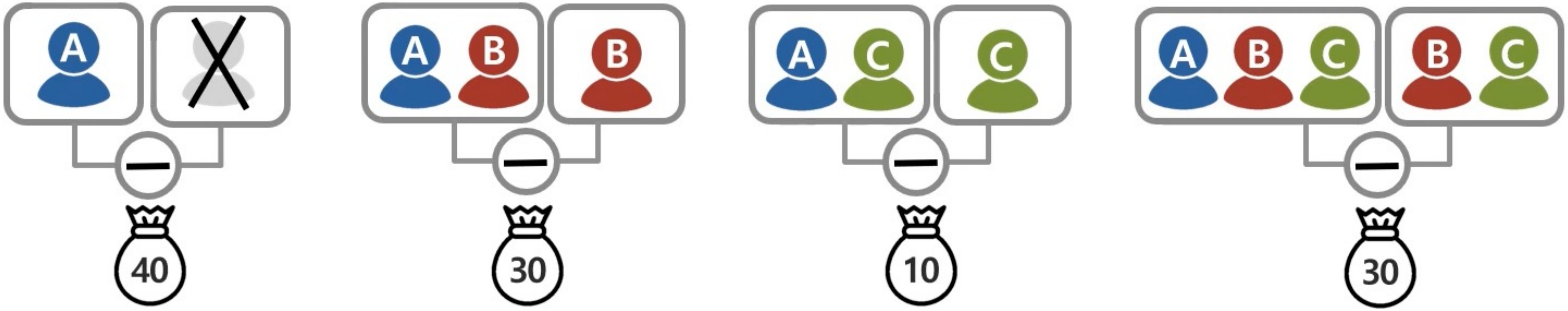
$$\frac{|S|!\,(|F|-|S|-1)!}{|F|!}$$

```
F = total player set
S = all subsets but target(Shapley Value) player
```

SHAP

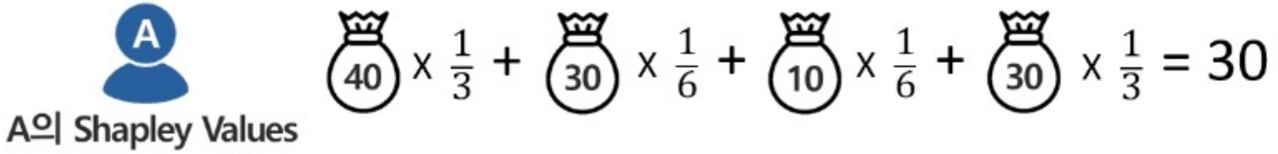
- Shapley Values
  - A weighted average of each pla… …itions
  - Marginal contribution of a play…
    “For each possible team(subset… payoff increases when **this player** joins that team.”

$F = \{$  $\}$

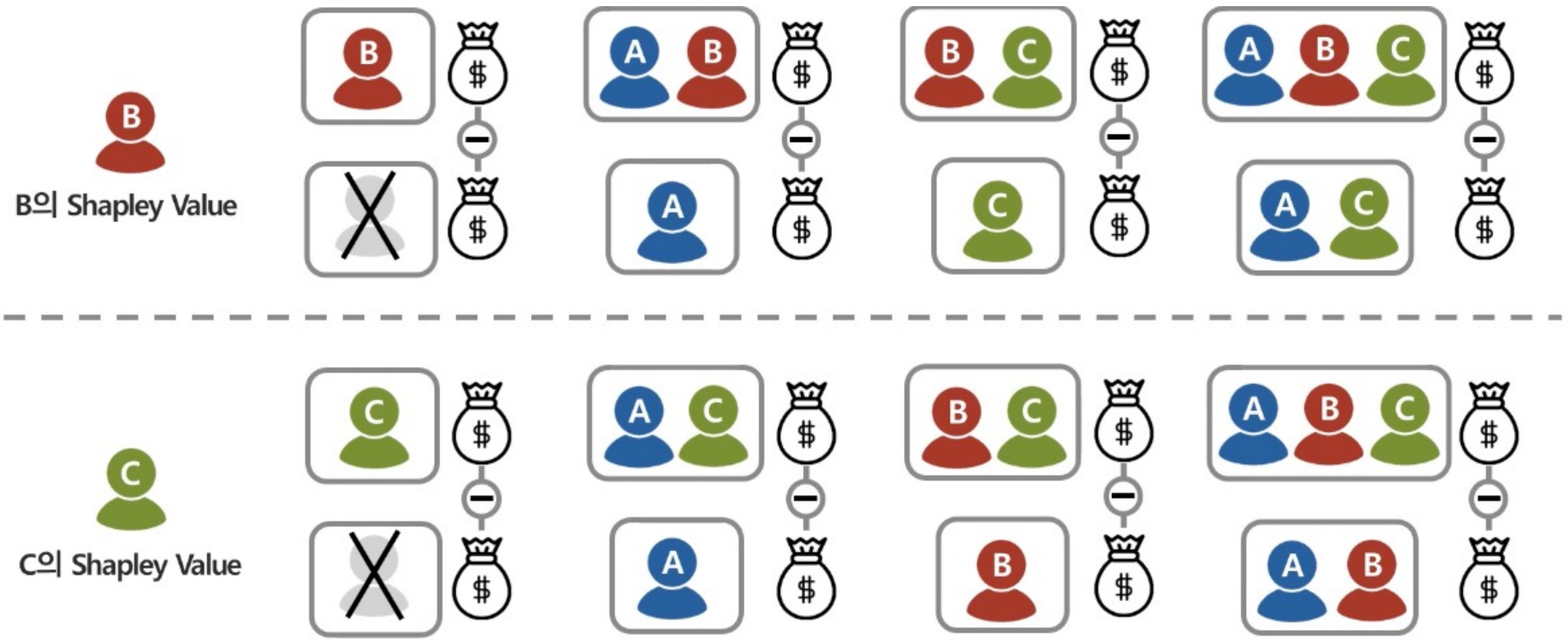


$$\frac{0!(3-0-1)!}{3!} = \frac{1}{3} \qquad \frac{1!(3-1-1)!}{3!} = \frac{1}{6} \qquad \frac{1!(3-1-1)!}{3!} = \frac{1}{6} \qquad \frac{2!(3-2-1)!}{3!} = \frac{1}{3}$$

$$\frac{|S|!\,(|F|-|S|-1)!}{|F|!}$$



SHAP


B
B의 Shapley Value
B
A B
B C
A B C
A
C
A C
C
C의 Shapley Value
C
A C
B C
A B C
A
B
A B

SHAP

- SHAP(Shapley Additive exPlantions)
  - Core contribution: using Shapley Values, calculate feature importance

Model’s input features

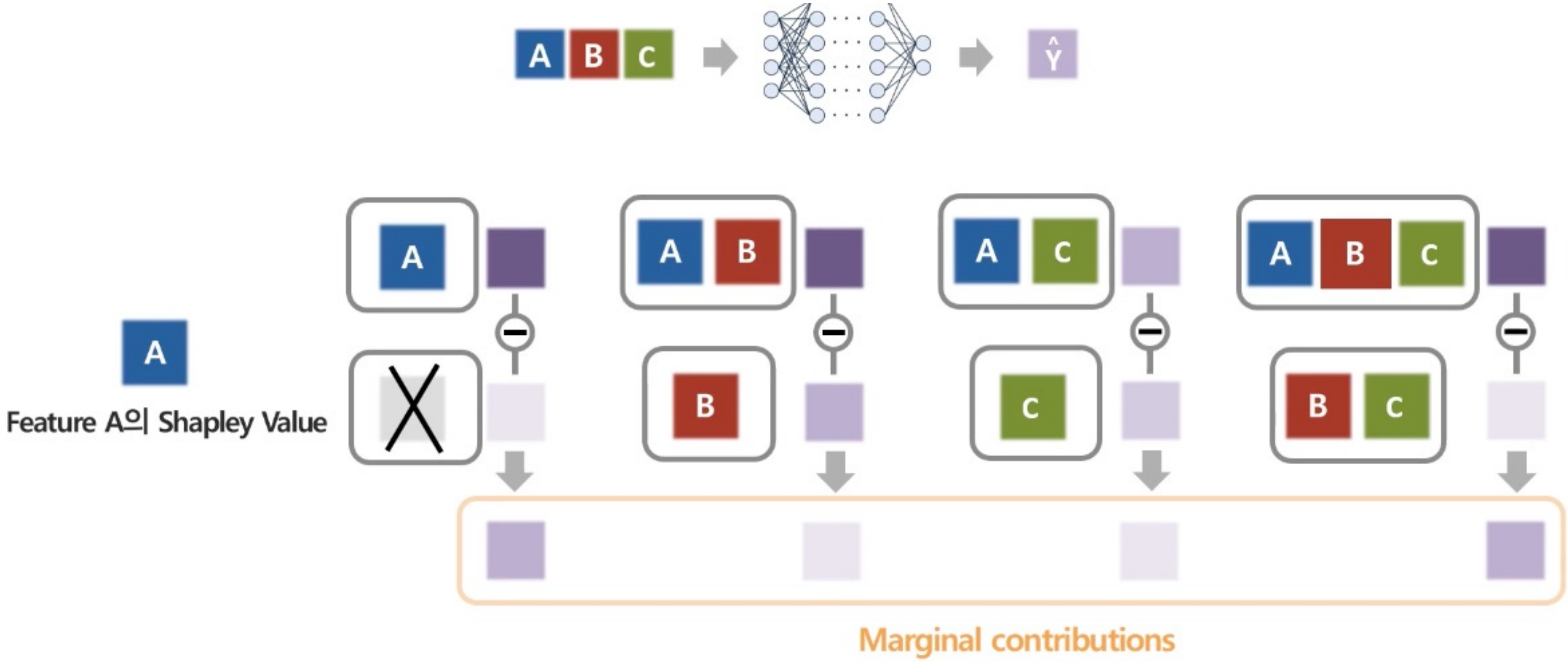
player

Model’s predictions

prize

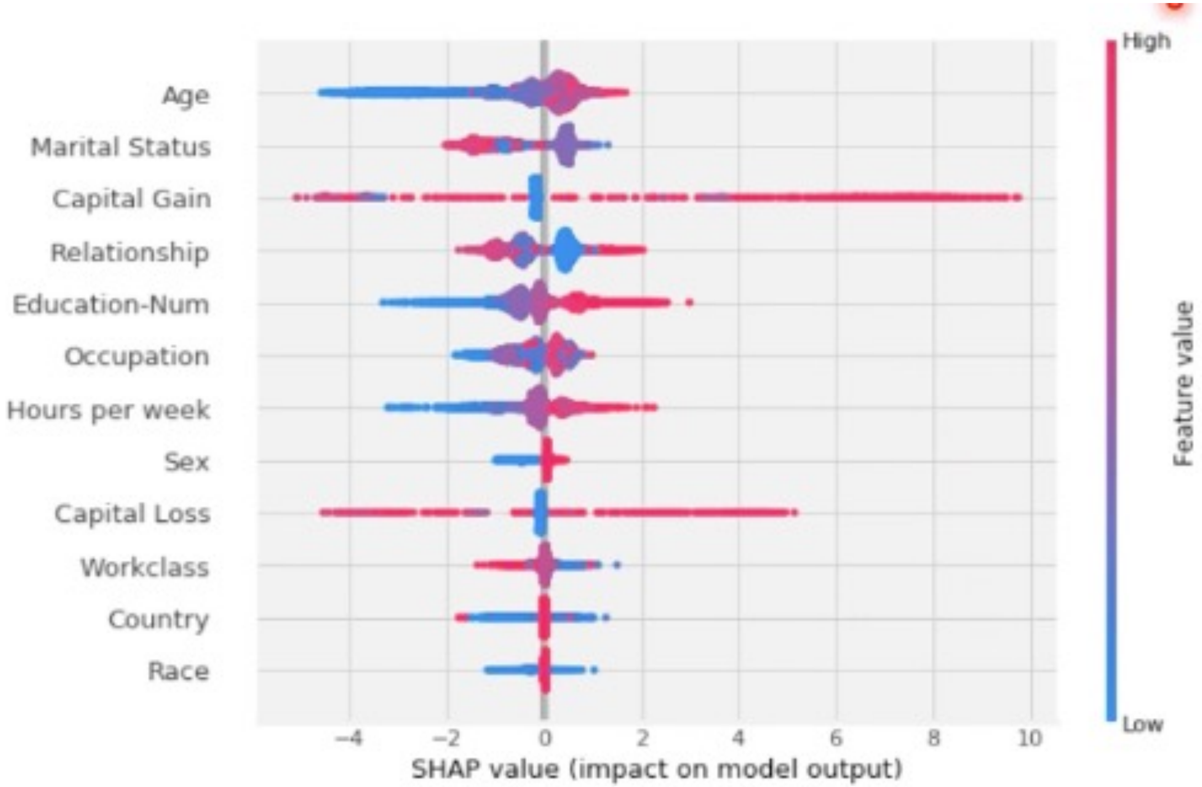
SHAP

- SHAP(Shapley Additive exPlantions)



SHAP

- SHAP(Shapley Additive exPlantions)
  - Binary Classification problem for Modern American income (50K $ up / down)
  - More age → more chance to get 50K$
  - Higher Education → more chance to get 50K$



Fin.

Thank you.

* 이 PPT는 아주대학교에서 제공한 아주체과 아모레퍼시픽의 아리따부리, consolas 글꼴를 사용하고 있습니다.